

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Comparative Analysis

Code of Conduct:

I __Harshavardhan. RV(192472163)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Harshavardhan. RV

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	

Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A 2.0 2.5

B 3.0 3.8

C 4.0 5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

ANS:

1. Depth Estimation Comparison: Stereo vs SfM

Given Data

Object	Actual Depth (m)	Stereo (m)	SfM (m)
A	2.0	2.0	2.5
B	3.0	3.0	3.8
C	4.0	4.0	5.0

(a) Absolute Error

Formula:

$$|\text{Estimated Depth} - \text{Actual Depth}|$$

Object A:

- Stereo = $|2.0 - 2.0| = 0 \text{ m}$
- SfM = $|2.5 - 2.0| = 0.5 \text{ m}$

Object B:

- Stereo = $|3.0 - 3.0| = 0 \text{ m}$
- SfM = $|3.8 - 3.0| = 0.8 \text{ m}$

Object C:

- Stereo = $|4.0 - 4.0| = 0 \text{ m}$
- SfM = $|5.0 - 4.0| = 1.0 \text{ m}$

Error Comparison

Object	Stereo Error	SfM Error
A	0 m	0.5 m
B	0 m	0.8 m
C	0 m	1.0 m
Total Error	0 m	2.3 m

(b) Comparison and Evaluation

Stereo Vision is more accurate because its error is **0 m** for all three objects.

- Stereo total absolute error = **0 m**
- SfM total absolute error = **2.3 m**

Therefore, based on the given data, **Stereo Vision performs significantly better than Structure from Motion (SfM)** for depth estimation.

Depth Estimation Error

Absolute error for Stereo Vision and Structure from Motion.

object	stereo	sfm
A	0	0.5
B	0	0.8
C	0	1

2. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional 85 100

Deep Learning 92 100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

Given Data

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Accuracy Calculation

Formula:

Traditional System:

Deep Learning System:

(b) Comparison

System	Accuracy
Traditional	85%
Deep Learning	92%

Difference:

Evaluation

The Deep Learning system performs better because it achieves 92% accuracy, compared with 85% for the Traditional system.

Therefore, Deep Learning improves accuracy by 7 percentage points and correctly recognizes 7 more samples out of every 100 than the traditional system.

Conclusion:

Deep learning is the better learning system

3. Precision-Recall Trade-off Analysis

Two models produce:

Model TP FP FN

A 70 30 20

B 85 25 25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

Given Data

Model	TP	FP	FN
A	70	30	20
B	85	25	25

(a) Precision and Recall

Formulas

$$Precision = \frac{TP}{TP + FP}$$
$$Recall = \frac{TP}{TP + FN}$$

Model A

Precision:70%

Recall:77.78%

Model B

Precision:77.27%.

Recall:.77.27%

(b) Comparison

Model	Precision	Recall
A	70%	77.78%
B	77.27%	77.27%

Model A has **higher recall**, but its precision is lower.

Model B has almost equal precision and recall, both at **77.27%**.

Evaluation

Model B is more balanced because its precision and recall are equal.

Therefore:

It provides a better trade-off between **avoiding false positives** and **detecting actual positive cases**.

4. F1-Score Comparison

Two models:

Model Precision Recall

A 0.8 0.6

B 0.75 0.75

(a) Compute F1-score for both models.

(b) Compare and evaluate which model is better.

4. F1-Score Comparison

Given Data

Model Precision Recall

A 0.8 0.6

B 0.75 0.75

(a) F1-Score Calculation

Formula:

F1-Score Formula

Model A

Given:

- Precision = 0.8
- Recall = 0.6

Substitute the values:

Model B

Given:

- Precision = 0.75
- Recall = 0.75

Final Answer

- Model A F1-score = 0.686
- Model B F1-score = 0.750

Therefore, Model B is better because $0.750 > 0.686$.

(b) Comparison

Model	Precision	Recall	F1-Score
A	0.80	0.60	0.686
B	0.75	0.75	0.750

Evaluation

Model B is better because it has a higher F1-score:

[
 $0.750 > 0.686$

]

Although Model A has higher precision, its recall is considerably lower. Model B provides a better **balance between precision and recall**.

Conclusion

[
 $\boxed{\text{Model B performs better based on F1-score.}}$

]

5. Optical Flow Consistency

Motion values:

Frame Method A Method B

1 4 6

2 5 8

3 6 9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

5. Optical Flow Consistency

Given Data

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute Average Motion

Formula:

Method A

Method B

(b) Compare Consistency

Method A:

Motion values are:

Changes between frames:

So, Method A changes consistently by **1 unit**.

Method B:

Motion values are:

Changes:

Method B has a larger variation between frames.

Evaluation

Method	Average Motion	Consistency
A	5.00	More consistent
B	7.67	Less consistent

Therefore, **Method A is more stable and consistent**, because its motion increases uniformly by **1 unit per**